

DAY-3 Assignments

Concepts: Class, Object, Constructors, Properties

Problem 1: Simple Calculator Using Methods

Scenario:

A small retail shop wants a simple calculator application to perform addition and subtraction operations using reusable methods.

Requirements:

1. Create a class named Calculator.
2. Create methods Add(int a, int b) and Subtract(int a, int b).
3. Each method should return the result.
4. In Main(), create an object and call the methods.
5. Display the output.

Technical Constraints:

1. Use method parameters and return types properly.
 2. Use appropriate access modifiers.
 3. No global variables allowed.
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Problem 2: Student Grade Calculator

Scenario:

A school wants to calculate the average marks of a student using a class-based approach.

Requirements:

1. Create a class Student.
2. Create method CalculateAverage(int m1, int m2, int m3).

3. Return the average marks.
4. Display grade based on average.

Technical Constraints:

1. Use return type double for average.
2. Avoid hard-coded values.

Expectations:

Clear separation of logic inside methods.

Learning Outcome:

Learn method creation, return values, and basic OOP concepts.

Sample Input:

80 70 90

Sample Output:

Average = 80, Grade = A

Problem:3 Write a C# program to process customer details:

Class Name: Customer

- Class should allow the following parameters in constructor

Customer c1 = new Customer();

Customer c2 = new Customer(10256);

Customer c3 = new Customer(10256, "Scott");

Customer c4 = new Customer(10256, "Scott", "Hyd");

- Declare required number of variables in the class.
- Method : ShowDetails() to display all the values in single Line.

Eg: Customer Details :: Id : 10256, Name : Scott, City : Hyd

Problem-4: Create a c# Program class to process product details:

- a. Details : product id, product name, unit price, quantity
- b. create required fields, properties
- c. perform read and write operations using properties
 - > read data from user and assign to properties
 - > display the details using properties

Eg: `Product obj = new Product();`
 `obj.ProductId = int.Parse(Console.ReadLine());`
 `Console.WriteLine("Product Id : " + obj.ProductId);`

Problem-5: Write a c# program to implement the following requirements:

Create an Employee Class

Members:

1. Properties: `Id, Name, Job, Basic, HRA, DA, IT, PF, GrossSalary`
2. Methods : `GetGrossSalary(), PrintDetails()`

GrossSalary should calculate based on the following following requirements :

$$\text{Gross Salary} = \text{Basic} + \text{HRA (15\%)} + \text{DA (8\%)} - \text{IT(10\%)} - \text{PF(5\%)}$$

- Read input from user : `Id, Name, Job, Basic`
- Make sure that all options are working without fail.
- Add extra properties to store `HRA, DA, IT, PF, GrossSalary` etc....